ASSIGNMENT 5 — QUESTION 9

ES6 Class Inheritance

JavaScript (ES6+) solution

```
class Person {
    constructor(name) {
        this.name = name;
    }

    showName() {
        console.log(`Name: ${this.name}`);
    }
}

class Student extends Person {
    constructor(name, branch) {
        super(name);
        this.branch = branch;
    }

    showBranch() {
        console.log(`Branch: ${this.branch}`);
    }
}

const student = new Student("Mohit", "CSE");
student.showName();
student.showBranch();

console.log(student instanceof Student); // true
console.log(student instanceof Person);  // true
```

Expected Output

```
Name: Mohit
Branch: CSE
true
true
```

Explanation

extends creates the inheritance relationship, and super(name) calls the parent Person constructor before this is used in Student. This class version provides the same basic behavior as the constructor-function prototype version in Question 5.